Geophys. J. Int. (1992) **108**, 883–894

Early Devonian (pre-Acadian) magnetization directions in Lower Old Red Sandstone of south Wales (UK)

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Accepted 1991 September 30. Received 1991 September 30; in original form 1991 May 8

SUMMARY

Two components of magnetization have been resolved from the Lower Old Red Sandstone of south Wales. All 39 sampled sites lie north of the Variscan front, and all but three are largely unaffected by Variscan folding. Thirteen sites yield a high unblocking temperature magnetization component which pre-dates Acadian (*mid*-Devonian) folding. The site mean direction for this component (Dec.: 232.2°, Inc.: 31.9°, α_{95}: 8.5°) yields a pole position (Lat.: 7.3°S, Long.: 306.7°E) which can be assigned an Early Devonian age. A lower unblocking temperature component is present at 38 of the sampled sites. This component post-dates both the Acadian *and* Variscan folding, has reversed polarity and a mean direction (Dec.: 193.8°, Inc.: −7.2°, α_{95}: 2.6°) which corresponds to a pole position (Lat.: 40.4°S, Long.: 338.0°E) consistent with a latest Carboniferous or early Permian age.

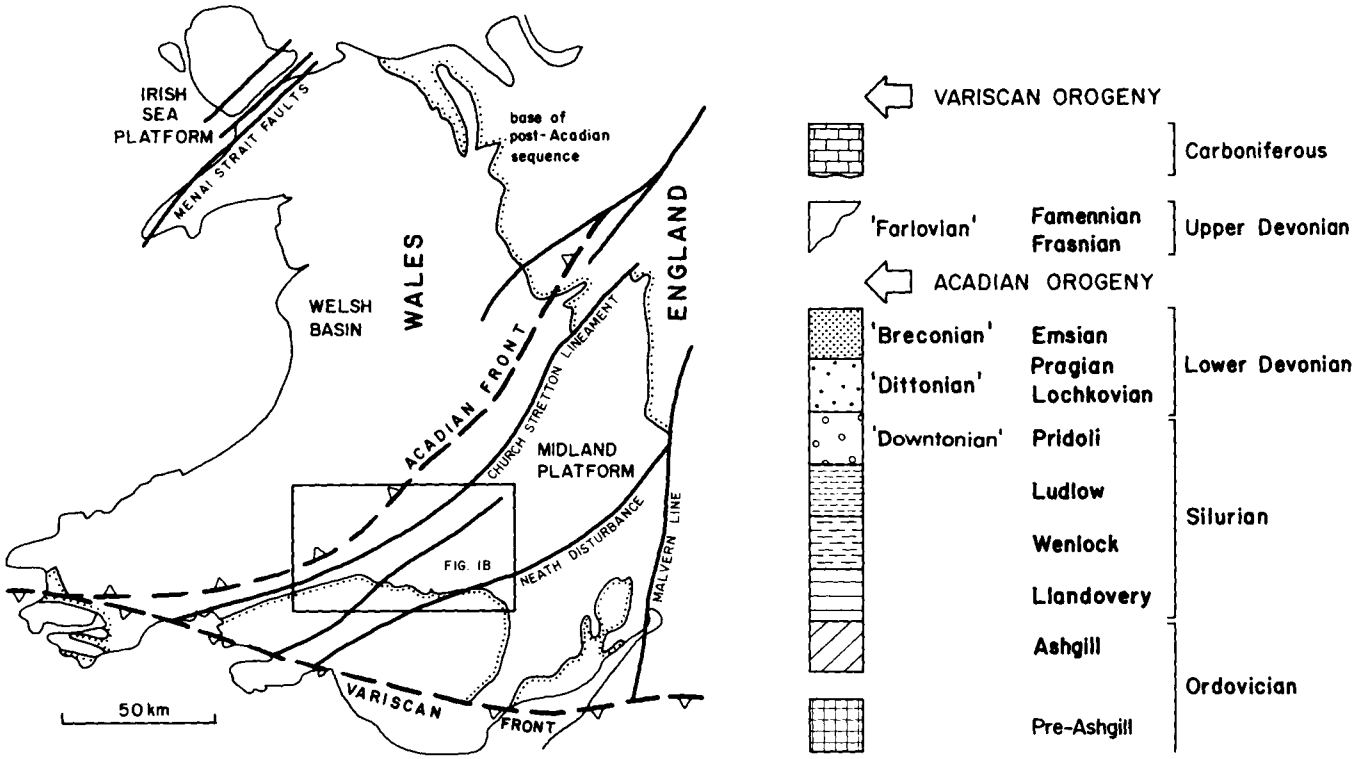
The Early Devonian pole position confirms the existence of a Siluro-Devonian inflexion in the apparent polar wander path (APWP) for southern Britain which coincides with an inflexion in the APWP for Scotland (Britain north of the Iapetus suture), indicating the closure of the Iapetus Ocean by Siluro-Devonian time. The Early Devonian palaeolatitude (17°S ± 5°) is consistent with an Early Devonian configuration in which Gondwana, Laurentia, Baltica and Avalonia were part of a single supercontinent.

Key words: Early Devonian, Old Red Sandstone, palaeomagnetism, Wales.

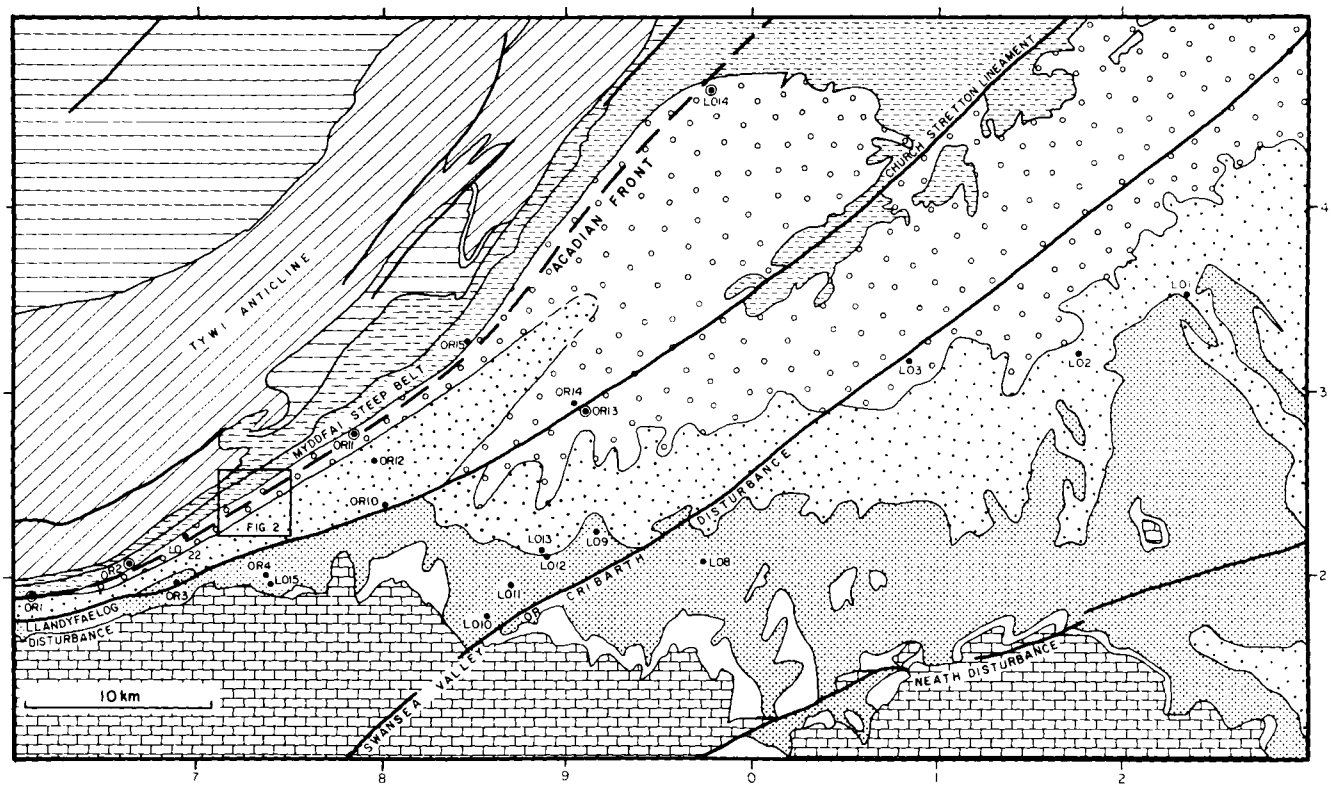
1 INTRODUCTION

The Old Red Sandstone of south Wales was the subject of a classic analysis of multicomponent magnetization in the early days of a palaeomagnetic research (Creer 1962; Chamalaun & Creer 1964; Chamalaun 1964). The high unblocking temperature mean direction (Dec.: 66°, Inc.: −37°) given by Chamalaun & Creer (1964) is quoted in all Palaeozoic apparant polar wander paths (APWP) for Britain south of the Iapetus suture (e.g., Briden & Duff 1981; Briden, Turnell & Watts 1984; Van der Voo 1988; Kent & Van der Voo 1990; Trench & Torsvik 1991). This pole position is particularly important to the APWP (and hence to the closure history of the Iapetus Ocean) due to the paucity of Siluro-Devonian palaeomagnetic data from southern Britain and to the fact that the pole defines an inflexion in the path. In the Chamalaun & Creer (1964) study, the sites are located mainly (but not entirely) within the Variscan belt, and although the presence of a pre-folding magnetization component was recognized, the component directions are not well resolved and the mean direction is poorly defined. Subsequent palaeomagnetic studies of the Old Red Sandstone from south Wales (McClelland Brown 1983; Stearns & Van der Voo 1987), using modern methods for resolving magnetization components, have yielded both pre-folding (Stearns & Van der Voo 1987) and synfolding (McClelland Brown 1983) magnetization components. The mean direction of the pre-folding component (Dec.: 196°, Inc.: −5°) gives a pole position (Lat.: 39°S; Long.: 334°E) which lies on the Late Carboniferous part of the APWP, and is therefore considered to be Late Carboniferous in age (Stearns & Van der Voo 1987). The synfolding magnetization directions must also be Late Carboniferous in age, but have more westerly declinations which have been interpreted to indicate a post-folding clockwise rotation of about 40°

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(a)



(b)

Figure 1. Location map showing (a) the position of the Acadian and Variscan fronts and (b) site locations (excluding those located in the Sawdde Gorge, Fig. 2), major fault zones and the basic geology. Circled site locations are those with resolvable pre-Acadian magnetization components.

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(McClelland Brown 1983). The two palaeomagnetic studies can be reconciled by postulating that the structures sampled by Stearns & Van der Voo (1987) are late structures which post-date the clockwise rotation (see McClelland 1987).

The need for Devonian control of the APWP for Britain south of the Iapetus suture (also known as Eastern Avalonia) led us to sample the Old Red Sandstone north of the Variscan front and to endeavour to resolve a magnetization component which pre-dates the *Acadian* deformation. Such a component would have tight age constraints, being restricted to the Ludlow to Emsian (Late Silurian–Early Devonian) interval.

2 GEOLOGICAL SETTING

39 sites in the Lower Old Red Sandstone (Upper Silurian to Lower Devonian) were sampled in central south Wales (Fig. 1). In this region and throughout much of Wales, Acadian (late Caledonian) deformation affects the dominantly marine sedimentary and volcanic rocks of the Early Palaeozoic Welsh Basin (Fig. 1). Coeval marine sediments of the Midland Platform are preserved southeast of the Acadian deformation front. Here they pass up conformably into the non-marine sediments of the Lower Old Red Sandstone (Pridoli to Emsian, House *et al.* 1977). An unconformity spanning at least mid-Devonian time marks the uplift and erosion associated with Acadian deformation. The platform area suffered only minor Acadian tilting, and a correspondingly weak angular unconformity occurs at the base of the post-Acadian sequence.

Non-marine Upper Old Red Sandstone (Frasnian and Famennian) is overlain by marine Lower Carboniferous and then by paralic Upper Carboniferous facies formed in a developing foreland basin to the Variscan belt to the south (Kelling 1988). The northern limit of the strong Variscan deformation (late Carboniferous to earliest Permian) runs E–W across the old Midland Platform, just overlapping the Acadian front in the west (Fig. 1).

The sampling area was chosen to lie north of the Variscan Front, but to overlap the Acadian Front, allowing fold tests that tightly constrain the age of the magnetization components. Particular attention was focused on a major monoclinal fold at the Acadian Front, comprising the southeast younging Myddfai Steep Belt and the low-dipping rocks of the platform to the SE (Fig. 1). Intensive sampling was carried out along the Sawdde Valley transect across this structure (Fig. 2). The fold is considered (Woodcock 1987) to be Acadian in age because: (a) it affects a conformable sequence up to 'Dittonian' (Upper Lochkovian to Pragian) but not Upper Old Red Sandstone or Carboniferous, and (b) it has a geometrically congruous cleavage, visible in Ludlow and older rocks, which is continuous with the main Acadian cleavage of central Wales.

The sampling area is cut by three NE–SW fault belts, the Church Stretton, Swansea Valley and Neath (Fig. 1), bordered sporadically by local upright folds. Followed southwest, these fault belts all cut Carboniferous rocks, and therefore have a component of Variscan displacement, probably sinistral strike-slip (Owen & Weaver 1983). They may overlie Precambrian basement features, and Acadian

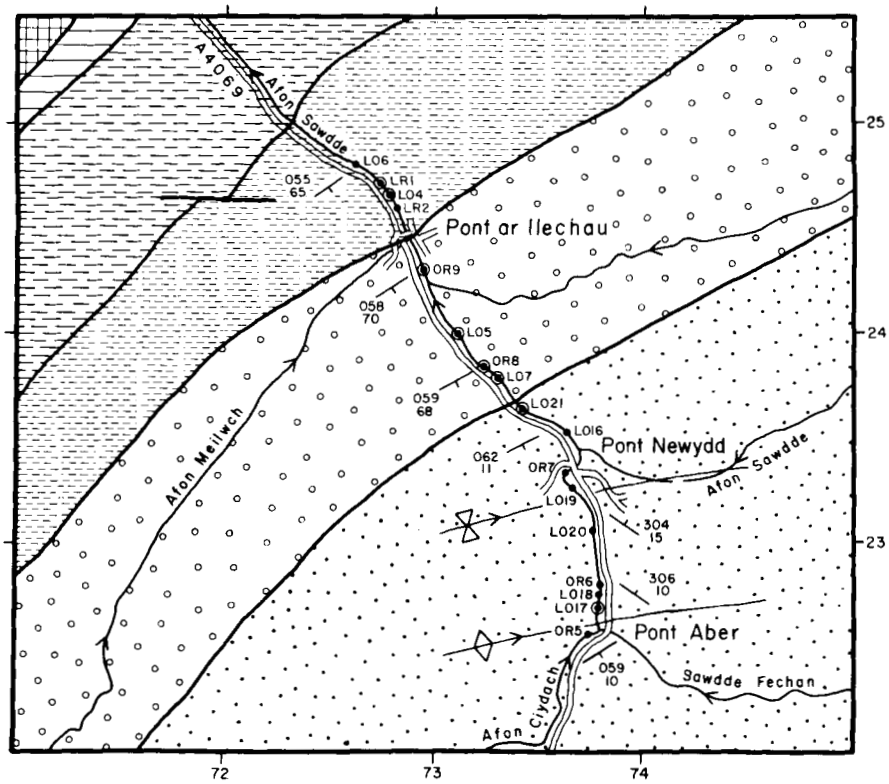


Figure 2. Geologic sketch of Sawdde Gorge, showing site locations. Stratigraphic ornament as for Fig. 1. Circled site locations are those with a resolvable pre-Acadian magnetization components. Grid markings are in 1 km intervals.

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displacement components on some or all of them cannot be ruled out. This ambiguity complicates structural interpretation at a few of the sites.

3 MAGNETIC ANALYSIS

Progressive thermal demagnetization of the Old Red Sandstone samples indicates the presence of two resolvable magnetization components (Figs 3 and 4). Component directions were determined by picking (by eye) the straight line segments from the orthogonal projections of the thermal demagnetization data, and using principle component analysis (Kirschvink 1980) to compute the component directions. The lower unblocking temperature component dominates the magnetization at 38 of the 39 sites. At one site (LO11), no resolvable magnetization component was observed. At 25 sites, the lower unblocking temperature component is the only resolvable magnetization component. It is characterized by a distributed unblocking temperature spectrum, and is demagnetized over a temperature range from a few hundred degrees centigrade to about 640 °C.

The site mean low unblocking temperature component directions are well grouped before structural tilt correction, and become dispersed after tilt correction (Fig. 5, Table 1), indicating that the component post-dates the *Acadian* folding. There is no evidence that the component is synfolding. Incremental unfolding indicates maximum precision at 0 per cent tilt correction. The polarity of this component is always reversed.

Three sites (LO10, LO12 and LO13, Fig. 1) are in folds adjacent to the Cribarth (Swansea Valley) disturbance, assumed to be of Variscan age. For these three sites, the component site mean directions again disperse on tilt correction indicating that this component also post-dates *Variscan* folding (Fig. 6, Table 2).

The high unblocking temperature component was resolvable at 13 sites (Table 3); at some other sites the higher unblocking temperature component was present but its direction was not resolvable. The high unblocking temperature component has a discrete unblocking temperature spectrum in the 640°–680 °C temperature range (Figs 3 and 4). The site mean high unblocking temperature component directions are better grouped *after* structural tilt

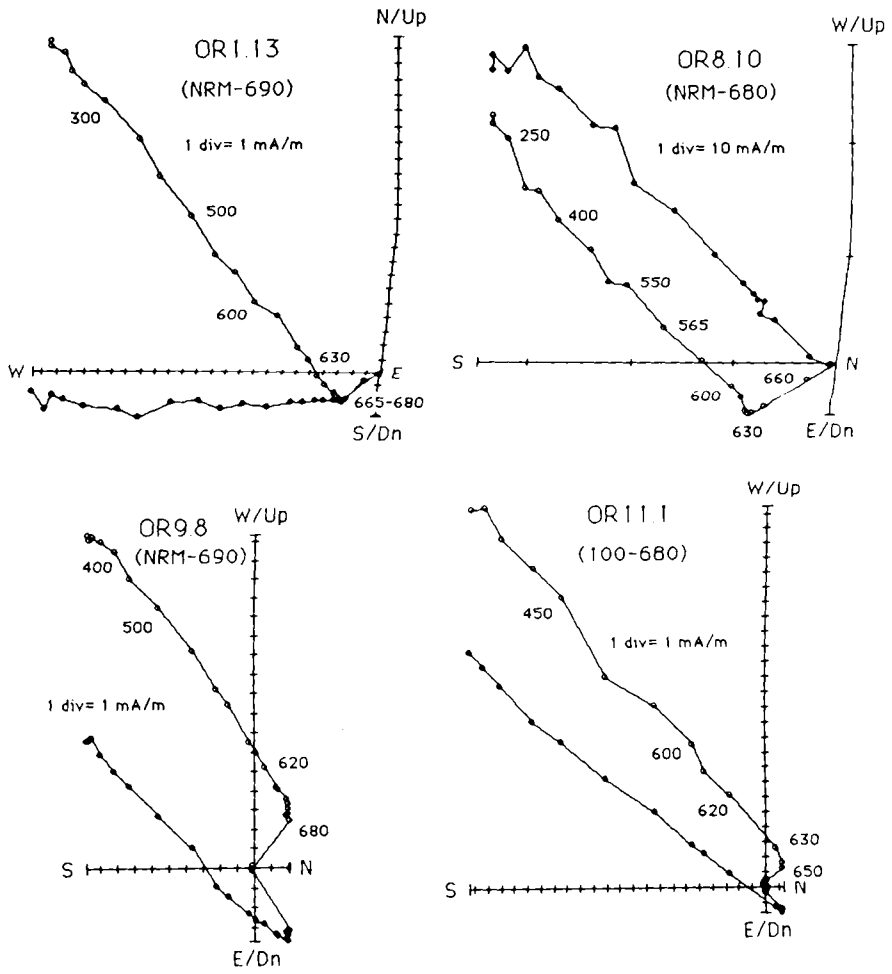


Figure 3. Orthogonal projections of thermal demagnetization data (with structural tilt correction) for samples from different Old Red Sandstone sites. Open and closed symbols indicate projection on the vertical and horizontal planes, respectively. The site and sample numbers, and demagnetization ranges are shown. Demagnetization temperatures (in degrees Celsius) are indicated for some vertical projection points.

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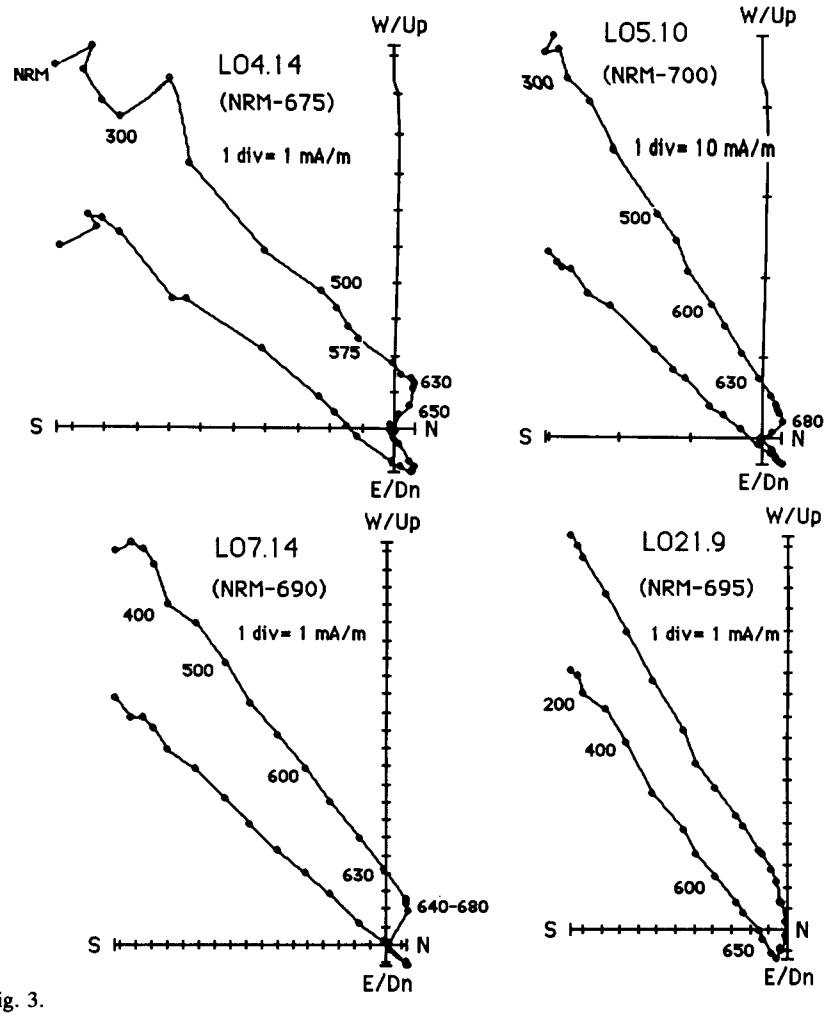


Figure 4. See caption to Fig. 3.

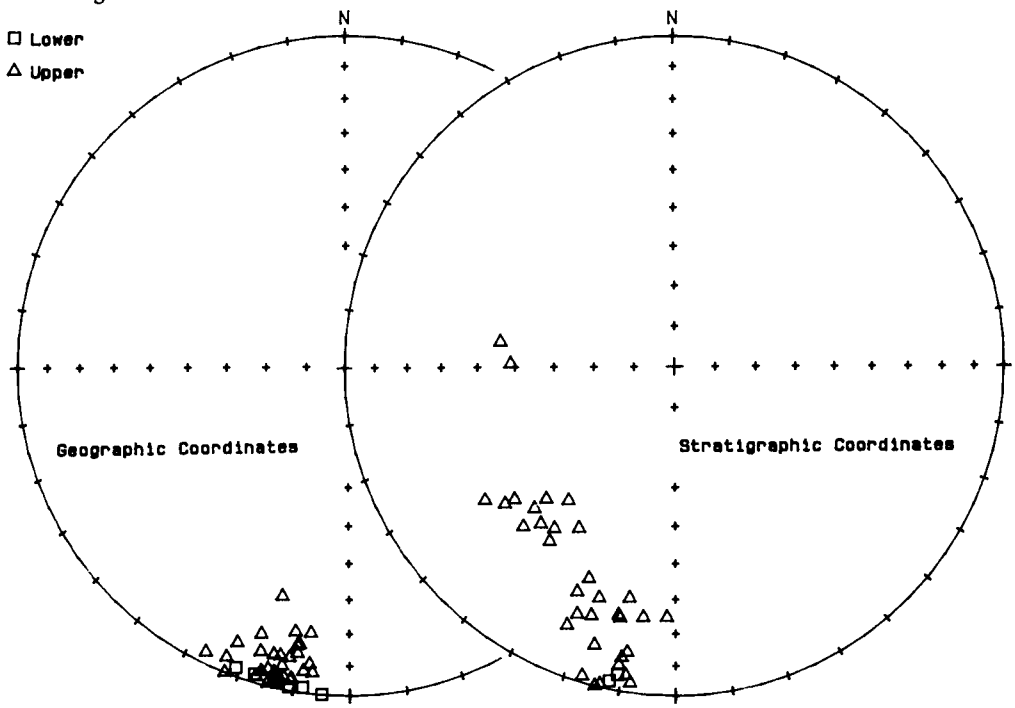


Figure 5. Equal area projection of low unblocking temperature component site mean directions before (left) and after (right) structural tilt correction, for sites located outside the zone of Variscan deformation and partly within Acadian structures. Squares and triangles represent positive (downward) and negative (upward) inclinations, respectively.

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Table 1. Low unblocking temperature component directions from sites located outside the Variscan deformation zone and partly within Acadian structures. N/No is the ratio of samples collected to samples for which a component was resolved. The parameters k and α_{95} refer to statistical parameters after Fisher (1953). The polar error (dp/dm) is the oval of 95 per cent confidence about the pole.

Site	N/No	In Situ		After	Tilt	k	α_{95}
		Dec.	Inc.	Dec.	Inc.		
LR1	16/16	193.4	-11.0	224.8	-39.4	180.9	2.7
LR2	12/15	188.4	2.0	210.8	-42.5	81.0	4.9
OR1	14/14	194.7	-10.9	271.2	-48.9	883.9	1.3
OR2	6/7	198.2	-16.0	278.2	-45.8	184.2	4.9
OR3	17/17	192.0	-4.3	181.9	-24.8	167.8	2.8
OR4	19/19	197.3	-10.7	198.7	-21.4	65.0	4.2
OR5	15/15	190.1	-15.6	192.4	-23.0	79.5	4.3
OR6	14/15	187.6	-9.9	188.2	-3.4	286.4	2.4
OR7	14/14	197.2	3.0	203.6	-26.5	140.2	3.4
OR8	11/15	202.3	-0.1	223.6	-33.7	379.6	2.3
OR9	12/12	193.0	-4.3	224.3	-43.5	99.9	4.4
OR10	15/16	195.1	-5.8	201.7	-20.4	357.4	2.0
OR11	14/14	192.7	-7.9	216.8	-38.8	216.3	2.7
OR12	14/15	200.8	3.0	202.8	-16.1	99.2	4.0
OR13	15/15	190.2	-12.8	189.2	-5.7	234.9	2.5
OR14	11/12	188.1	-19.6	187.3	-24.3	248.8	2.9
OR15	14/14	184.9	0.2	198.3	-27.1	636.2	1.6
LO1	11/11	188.6	-7.4	189.6	-13.1	318.5	2.6
LO2	13/13	191.4	-19.2	191.2	-29.2	139.9	3.5
LO3	13/13	192.9	-1.6	193.5	-1.1	125.4	3.7
LO4	10/15	193.7	-2.2	215.8	-35.2	448.7	2.3
LO5	17/17	186.9	-7.2	218.6	-47.1	278.7	2.1
LO6	12/12	202.1	-10.9	231.2	-34.3	17.4	10.7
LO7	13/13	196.2	-4.5	220.6	-37.5	505.9	1.8
LO8	12/15	196.1	-28.5	202.2	-31.3	12.4	12.9
LO9	18/20	190.6	-4.3	190.5	-11.2	439.0	1.7
LO14	18/18	194.6	-1.1	196.3	-13.1	71.7	4.1
LO15	10/12	190.8	1.3	190.8	-8.7	64.6	6.1
LO16	15/15	190.5	-15.2	192.9	-23.7	147.7	3.2
LO17	16/16	193.9	-5.2	194.3	-0.2	237.6	2.4
LO18	13/14	190.7	-3.6	190.7	5.5	479.9	1.9
LO19	16/16	196.9	-2.3	196.9	-2.3	443.4	1.8
LO20	17/18	191.6	-11.0	192.0	2.9	431.9	1.7
LO21	10/12	203.0	-5.2	230.5	-37.1	62.0	6.2
LO22	13/13	206.8	-3.9	235.0	-30.4	180.8	3.1

MEAN Before tilt correction Dec: 193.8 Inc: -7.2 k=87.6 α_{95}=2.6
After tilt correction Dec: 204.8 Inc:-25.2 k=11.6 α_{95}=7.5
POLE Latitude: 40.4 S Longitude 338.0 E dp/dm= 3.8/7.5

Table 2. Low unblocking temperature component directions from sites in Variscan structures (symbols as for Table 1).

Site	N/No	In Situ		After	Tilt	k	α_{95}
		Dec.	Inc.	Dec.	Inc.		
LO10	16/16	194.8	4.1	197.9	-12.5	199.6	2.0
LO12	15/15	187.5	2.7	187.5	-4.9	28.0	6.3
LO13	11/15	185.6	10.6	198.5	31.4	65.4	5.7

MEAN Before tilt correction Dec: 189.3 Inc: 5.8 k= 159.9 α_{95}=9.8
After tilt correction Dec: 194.4 Inc: 4.5 k= 11.3 α_{95}=38.5

correction (Fig. 7, Table 3). Using the McElhinny (1964) fold test, the high unblocking temperature site mean magnetization directions pass the fold test at the 95 per cent confidence level, implying that the component pre-dates folding. Again, there is no evidence that this component is synfolding. Incremental unfolding indicates maximum precision at 100 per cent tilt correction. The depositional age of the Old Red Sandstone sites ranges from Ludlow to Emsian (latest Silurian to Early Devonian) (House *et al.* 1977). The Acadian folding is considered to be late Early to Mid-Devonian in age (Soper, Webb & Woodcock 1987; Woodcock *et al.* 1988). The high unblocking temperature magnetization component therefore has a Late Silurian–Early Devonian age. It is noteworthy that the sites carrying the higher unblocking temperature component are those located furthest from the Variscan front (Figs 1 and 2). They are also the stratigraphically lower sites. Since the proportion of sandstone in the Lower Old Red Sandstone broadly increases upwards (House *et al.* 1977), this may indicate a permeability control on the preservation of the older high unblocking temperature magnetization component.

4 ORIGIN OF THE MAGNETIZATION COMPONENTS

The thermal demagnetization behaviour of the Old Red Sandstone is reminiscent of the behaviour of Appalachian Palaeozoic redbeds (Kent & Opdyke 1985; Miller & Kent

Table 3. High unblocking temperature magnetization components from sites located outside the Variscan deformation zone and partly within Acadian structures (symbols as for Table 1). Lithostratigraphy after Almond (1983) and British Geological Survey (1988, 1990). Gwynfe Formation approximately equates to Raglan Mudstone Formation.

SITE ID	Lithostratigraphic Unit	N/No	In Situ		After	Tilt	k	α_{95}
			Dec.	Inc.	Dec.	Inc.		
LR1	Trichrug Fm.	12/16	266.9	9.0	239.6	35.2	27.3	8.5
OR1	Tilestones Fm.	10/14	277.0	4.4	238.6	29.2	82.2	5.4
OR2	Tilestones Fm.	6/7	280.3	12.8	231.4	29.1	35.0	11.5
OR8	Gwynfe Fm.	10/15	251.0	31.8	212.1	21.1	87.6	5.2
OR9	Gwynfe Fm.	12/12	88.2	-11.4	56.8	-32.1	68.8	5.3
OR11	Raglan Mudstone Fm.	14/14	88.9	-13.3	61.2	-36.4	51.2	5.6
OR13	Raglan Mudstone Fm.	14/15	60.9	-43.6	69.4	-43.1	140.7	3.4
LO4	Trichrug Fm.	9/15	275.6	8.7	244.0	43.0	43.3	7.9
LO5	Gwynfe Fm.	16/17	76.5	-14.9	50.3	-24.6	90.3	3.9
LO7	Gwynfe Fm.	9/13	101.0	-22.9	50.5	-49.1	21.9	11.3
LO14	Raglan Mudstone Fm.	11/18	211.7	28.1	204.7	19.3	22.0	9.9
LO17	Llanddeusant Fm.	10/16	215.3	10.3	213.6	18.0	28.3	9.2
LO21	Llanddeusant Fm.	5/12	267.9	-3.9	255.6	23.2	62.8	9.7

MEAN Before tilt correction **Dec:** 260.0 **Inc:** 17.0 k= 10.7 α_{95}= 13.3
After tilt correction **Dec:** 232.2 **Inc:** 31.9 k= 24.6 α_{95}= 8.5

POLE **Latitude:** 7.3 S **Longitude:** 306.7 E dp/dm= 5.4/9.5

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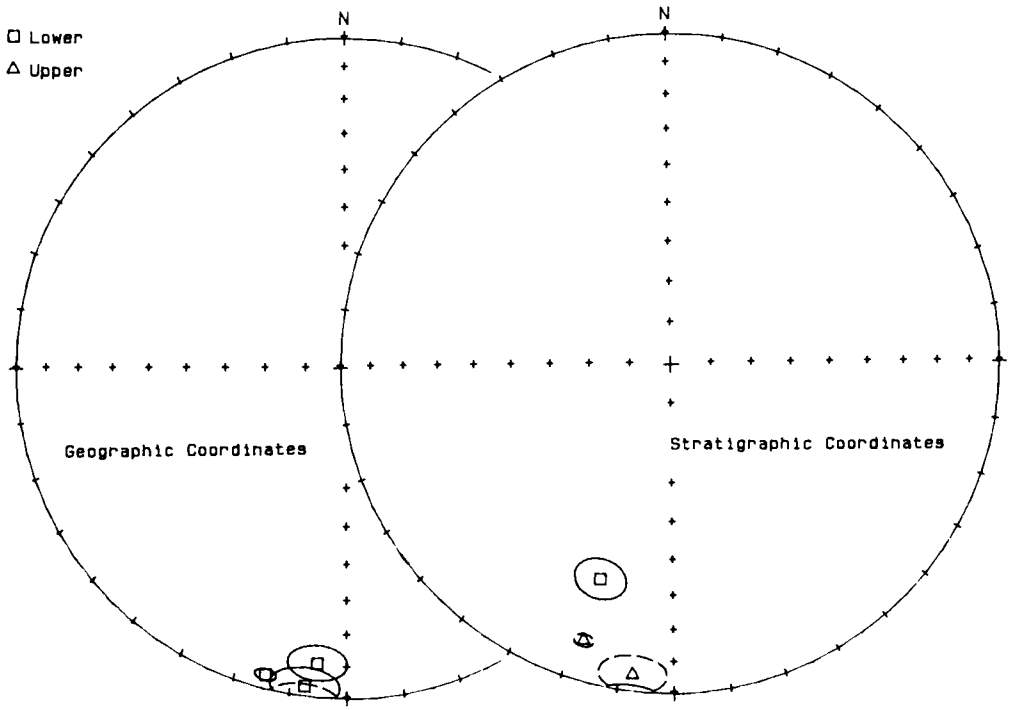


Figure 6. Equal area projection of low unblocking temperature component site mean directions before (left) and after (right) structural tilt correction, for sites located in Variscan structures. Circles of 95 per cent confidence are indicated (symbols as for Fig. 5).

1986, 1988). A lower unblocking temperature component dominates the magnetization and is characterized by dispersed unblocking temperatures in the range from a few hundred degrees Celsius to over 600 °C. A higher unblocking temperature component is occasionally observed and is characterized by a very discrete unblocking temperature spectrum in the 640°–680 °C temperature range.

In the Old Red Sandstone, the lower unblocking temperature component dominates the magnetization and was well resolved at all but one of the 39 sampling sites. This component clearly post-dates the Acadian folding (Fig.

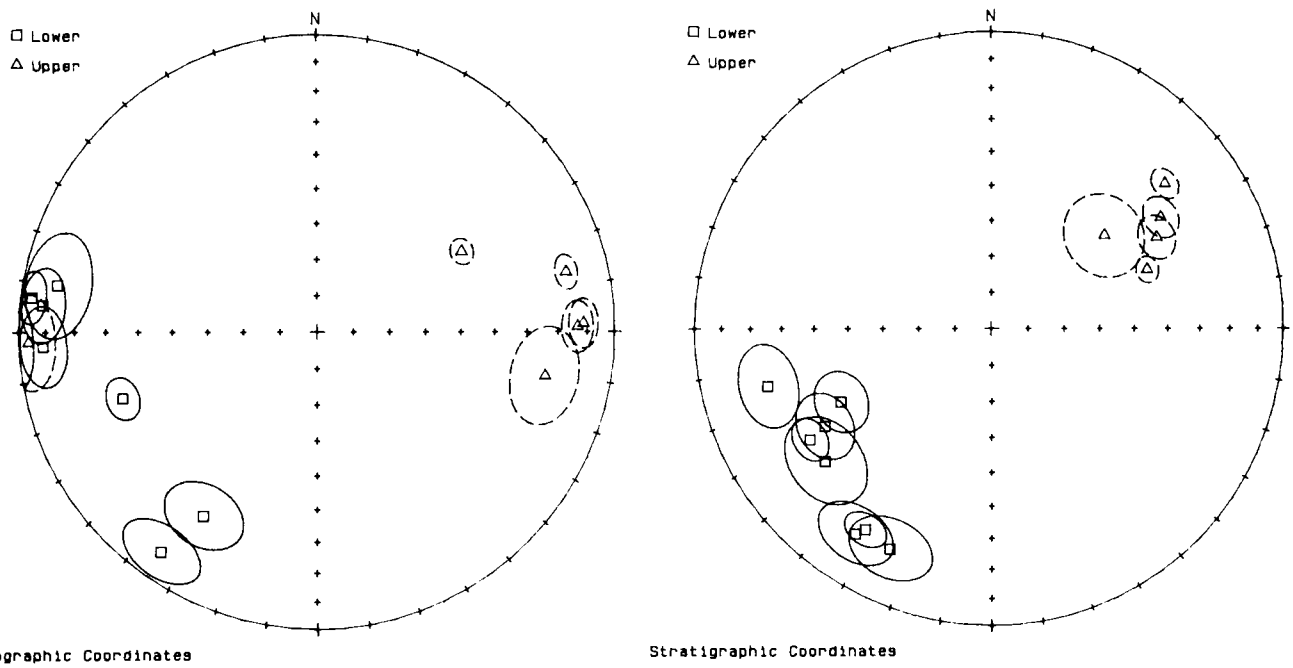


Figure 7. Equal area projection of high unblocking temperature component site mean directions before (left) and after (right) structural tilt correction, for sites located outside the zone of Variscan deformation and partly within Acadian structures. Circles of 95 per cent confidence are indicated (symbols as for Fig. 5).

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5, Table 1). The overall mean magnetization direction of the lower unblocking temperature component before tilt correction (Dec.: 193.8°, Inc.: −7.2°) is almost the same as, and probably coeval with, the *pre-folding* magnetization direction determined by Stearns & Van der Voo (1987) from the Variscan belt. The mean direction gives a pole position which implies a Late Carboniferous age.

Synfolding magnetizations and magnetizations coeval with adjacent compressional tectonics are common in North American Palaeozoic redbeds and it is generally considered (see McCabe & Elmore 1989) that these magnetizations are chemical remanences due to the growth of authigenic haematite mediated by fluid migration associated with the compressional tectonics. In view of the proximity of our sites to the Variscan front (Fig. 1), it is possible that the lower unblocking temperature component which we observe is a chemical remanence produced as a result of fluid migration associated with Variscan thrusting. Regional southward dips and the development of a late Carboniferous foreland basin (Kelling 1988) were a result of lithospheric loading by Variscan thrust sheets to the south. The fluids which may have promoted haematite growth could have migrated northwards, up dip in the sandstone aquifers of the Old Red Sandstone. These acquifers are more abundant in the upper part of the Lower Old Red Sandstone, perhaps explaining its more pervasive remagnetization. The lower unblocking temperature component always has reversed polarity and the mean direction gives a pole position (Lat.: 40.4°S, Long.: 338.0°E) which lies on the Late Carboniferous segment of the polar wander path for southern Britain (Fig. 8).

The high unblocking temperature pre-folding magnetization may possibly be a depositional remanence (DRM), or a chemical remanence produced during an early phase of authigenic haematite growth. The higher unblocking temperature implies a larger grain size of the authigenic haematite relative to that carrying the lower unblocking temperature component. The high unblocking temperature component exhibits dual polarity (Fig. 7) and passes the fold test (McElhinny 1964) at the 95 per cent confidence level (Table 3).

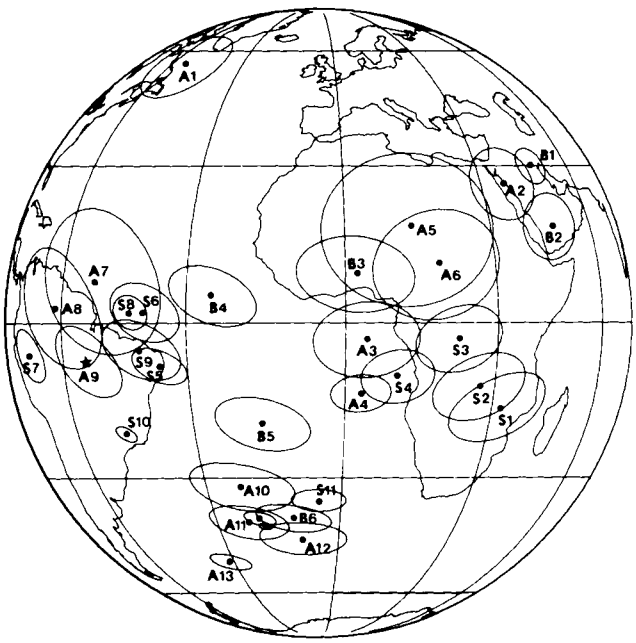


Figure 8. Selected Palaeozoic pole positions for southern Britain (prefix A), Scotland (prefix S) and Baltica (prefix B). The Early Devonian pole from this study is marked by a star. For index to poles, see Table 4.

5 APPARENT POLAR WANDER PATHS AND PALAEOGEOGRAPHY

The Palaeozoic palaeogeography of the Atlantic bordering continents and the closure history of the Iapetus Ocean has undergone considerable revision in the last few years. In the Early Ordovician, the 'Appalachian' margin of Laurentia was at low southerly latitudes, and the north African margin of Gondwana lay close to the south pole with a wide intervening Iapetus Ocean (see Van der Voo 1988). Ordovician palaeomagnetic data from Avalonian terranes of North America (Van der Voo & Johnson 1985), Armorica (Perround & Bonhommet 1981; Perround & Van der Voo 1985), southern Britain (Eastern Avalonia) (McCabe & Channell 1990; Trench *et al.* 1992) and Baltica (Claesson 1978; Torsvik *et al.* 1990) indicate that these continental units were at high or mid-southerly latitudes during Ordovician.

For the Late Ordovican to Late Silurian interval, prior to the Late Silurian/Early Devonian closure of Iapetus, the palaeogeography remains more uncertain, and the chronology of closure is not well constrained. It appears that a wide (about 30° latitudinal width) Early Ordovician Iapetus Ocean between Laurentia and Eastern Avalonia closed rapidly in Late Ordovician/Early Silurian time. One of the key questions is whether the Armorica, Eastern Avalonia and Baltica continental units were part of the northern rim of Gondwana during the Late Ordovician/Early Silurian closure, or whether the closure of an intervening ocean bordering Gondwana was a later (Hercynian) event. Unfortunately, there is a paucity of well-constrained palaeomagnetic data for the Late Ordovician to Late Silurian interval from Armorica, Eastern Avalonia and the Gondwana continents. Available data from Gondwana have been summarized by Bachtadse & Briden (1990). The Early Silurian African data of Hargraves, Dawson & Van Houten (1987) imply that the northern margin of West Gondwana moved rapidly to lower latitudes in latest Ordovician/Early Silurian time with the pole moving from NW Africa to the region off the SW tip of Africa (Fig. 9). Several Devonian poles from the Lachlan Fold Belt of SE Australia support this southerly pole position (Schmidt *et al.* 1986). Late Devonian Gondwana poles lie in central Africa, with the Late Devonian/Carboniferous poles making a well-defined southward path through central and southern Africa (Fig. 9). The Silurian 'excursion' of the Gondwanan APWP (first proposed by Morel & Irving 1978), from the Ordovican poles in North Africa to the Silurian poles off SW Africa to the Late Devonian poles in central Africa, implies that Gondwana moved northward with Armorica/East Avalonia/Baltica during Late Ordovician and Silurian time. The APWP for Gondwana from the Silurian poles off SW Africa to the Late Devonian poles located in central Africa would then denote the opening of a Late Devonian ocean

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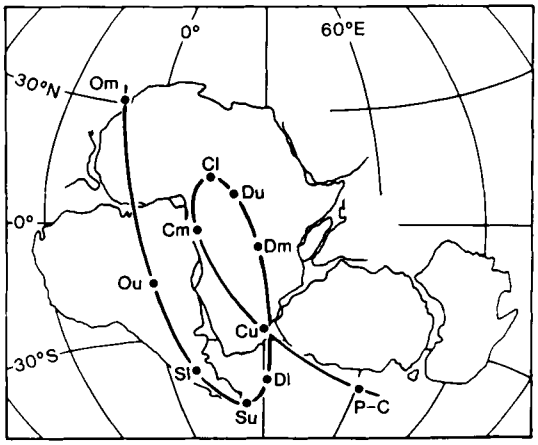


Figure 9. Apparent polar wander path for Africa from middle Ordovician (Om) to Permo-Carboniferous (P-C) time (after Bachtadse & Briden 1990).

between Gondwana and Armorica/Avalonia/Baltica, which subsequently closed during Hercynian orogeny.

Selected palaeomagnetic poles for Eastern Avalonia/Scotland/Baltica (the Old Red Continent) form three groups: Ordovician, Siluro-Devonian and Carboniferous/Permian (Fig. 8). The groupings imply either rapid apparent polar wander (continental drift) in the intervening time intervals or preferential magnetization of studied rocks at these times. The Ordovician to Siluro-Devonian interval is probably coeval with the rapid initial closure of the Iapetus Ocean. The Siluro-Devonian to Early Carboniferous interval may be coeval with the opening and closing of the Devonian-Carboniferous ocean which separated Gondwana from the Old Red Continent in the time interval between the Acadian and Hercynian orogenies. On the other hand, the separation of these two groups of poles may be an artifact of widespread remagnetization in Late Carboniferous/Permian time.

The Ordovician poles from Baltica generally plot close to coeval poles from Eastern Avalonia (Fig. 8) suggesting that Baltica and Eastern Avalonia had similar latitudes since Ordovician time. However, a pre-Variscan pole position from the Siluro-Devonian Ringerike Old Red Sandstone of southern Norway (Douglass 1988) (B5 in Fig. 8) is significantly different from the British Siluro-Devonian poles. The Ordovician pole positions from Scotland are generally different from the poles from Eastern Avalonia and Baltica. The two poles from the southern part of the Builth Inlier (A3 and A4 in Fig. 8) are exceptions and lie close to the Scottish poles. We believe that these two poles do not represent the Ordovician field due to problems of volcano-tectonic tilting in SW Builth (see McCabe, Channell & Woodcock, 1991; Channell *et al.* 1992).

For the Old Red Sandstone from Wales, the mean direction after tilt correction of the high unblocking temperature component (Dec.: 232.2°, Inc.: 31.9°) yields a pole position (Lat.: 7.3°S, Long.: 306.7°E) which lies near the Early Devonian segment of recently published APWPs for Eastern Avalonia (e.g. Trench & Torsvik 1991). This is because the pole is close to those from early studies of the Old Red Sandstone (Chamalaun & Creer 1964) and volcanics in Somerset and Gloucestershire (Piper 1975), on which the Siluro-Devonian inflexion of the published APWPs has been based (Fig. 8, Table 4). Both these early studies were carried out prior to the advent of modern methods of component analysis and were devoid of statistical fold tests. Nonetheless, the inflexion in the APWP which is defined by these two studies is essentially ratified by the new result (Fig. 8). This new Lower Old Red Sandstone pole is also very close to a number of Late Silurian to Early Devonian palaeomagnetic poles from Scotland (Fig. 8). The Siluro-Devonian inflexion is apparent in the APWPs from both the Scottish Midland Valley and the Grampian highlands (see Trench *et al.* 1989) (Fig. 8), suggesting that the present juxtaposition of major tectonic units in Britain was attained by Siluro-Devonian boundary time.

Table 4. Selected palaeomagnetic poles from southern Britain (Eastern Avalonia), Baltica and Scotland plotted in Fig. 8.

	Age	Pole Lat.	Pole Long.	dp/dm
SCOTLAND				
S1: Aberdeen Gabbro (Watts & Briden, 1984)	L. Ord	15.6 S	32.2 E	5.7/10.6
S2: Ballantrae Serpentinite (Piper, 1978)	L. Ord	12 S	27 E	6/10
S3: Dalradian Schist (Watts, 1985)	m. Ord	3 S	22 E	5.7/9.0
S4: Borrolan Ledmorite (Turnell & Briden, 1983)	L. Sil	10S	9.7E	5/7.3
S5: Arrochar Complex (Briden, 1970)	U.Sil/L.Dev	8 S	324 E	3.4/6.1
S6: Lorne Plateau (Latham & Briden,1975)	U.Sil/L.Dev	2 N	321 E	5/8
S7: Comrie Intrusion (Turnell, 1985)	U.Sil/L.Dev	6 S	287 E	4/7
S8: Strathmore Lavas (Torsvik, 1985b)	U.Sil/L.Dev	2.2N	318.2E	2.7/4.3
S9: Midland Valley Lavas (Sallomy & Piper, 1973)	U.Sil/L.Dev	5 S	320 E	5/9
S10: Esha Ness (Storetvedt & Torsvik, 1985)	U.Dev	21 S	314 E	1.3/2.5
S11: Argyllshire Dikes (Esang & Piper, 1984b)	m. Carb	35S	355 E	2.5/5
S12: Peterhead Dyke (Torsvik, 1985a)	Perm-Carb	41 S	342 E	0.7/1.3
BALTICA				
B1: Orthoceras Limestone (Claesson, 1978)	L. Ord	30 N	46 E	3.7/3.9
B2: Orthoceras Limestone (Torsvik and Trench, 1991)	L-M. Ord	18 N	46 E	6/8
B3: Pole B	U.Ord/Sil?	9 N	2 E	7/11
B4: Pole A (Torsvik et al., 1990)	M. Sil	5 N	335 E	5/9
B5: Ringerike Sandstone (Douglass, 1988)	U.Sil/L.Dev	19 S	344 E	4.8/9.3
B6: Skaane quartz-dolerites (Bylund, 1974)	U. Carb	39 S	349 E	3.4/6.7
B7: Ny Hellesund sills (Halvorsen, 1970)	U.C/L.Perm	39 S	341 E	1.5/3
SOUTHERN BRITAIN (Eastern Avalonia)				
A1: Treffgarne Volcanics (Trench et al., in press)	L. Ord.	56N	306E	9/10
A2: Stapeley Volcanics (McCabe and Channell, 1990)	m. Ord	26.6 N	36.1 E	6.9/8.2
A3: S. Builth volcanic/intrusion (Trench et al., 1991)	m. Ord	3 S	4 E	7/10
A4: S. Builth volcanic/intrusion (McCabe et al., in press)	m.Ord.	12.8S	3.2E	3.5/5.6
A5: N. Builth seds/volc/intr. (McCabe et al., in press)	m.Ord	17.9N	12.9E	14.7/17.2
A6: Tramore Volcanics (Deutsch, 1980)	m. Ord	11 N	18 E	10/13
A7: Somerset/Glouc. volcanics (Piper, 1975)	Sil	8 N	309 E	12.2/17.2
A8: Old Red Sandstone (Chamalaun and Creer, 1964)	U.Sil/L.Dev	3 N	298 E	9.2/15.1
A9: Old Red Sandstone (This paper)	U.Sil/L.Dev	7.3S	306.7E	5.4/9.5
A10: Portishead Red Beds (Morris et al., 1973)	L. Carb	32 S	338 E	5/10
A11: Old Red Sandstone overprint (This paper)	U.C/L.Perm	40.4 S	338 E	3.8/7.5
A12: Builth overprint (Trench et al., 1991)	U.C/L.Perm	44 S	351 E	4/8
A13: Exeter Lavas (Zijderveld, 1967)	L. Perm	50 S	329 E	2/4.3

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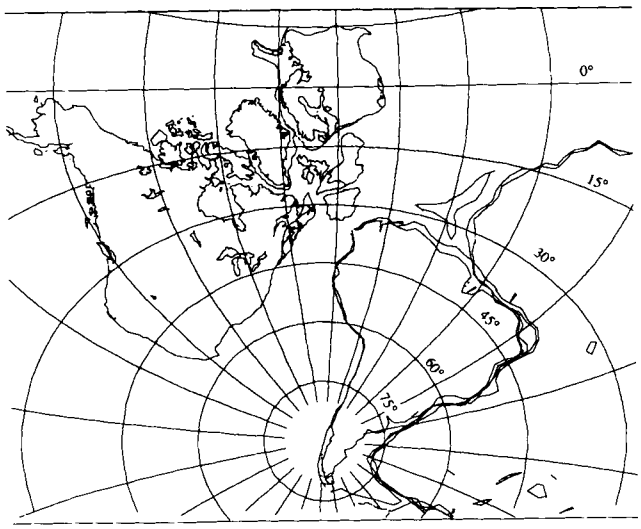


Figure 10. Siluro-Devonian reconstruction of the Old Red Continent (Laurentia, Avalonia, Baltica) and Gondwana. The Old Red Continent was assembled using the finite rotations of Bullard, Everett & Smith (1965) and positioned using (1) the mean Siluro-Devonian palaeomagnetic pole for North America (Van der Voo 1990) with $Q \geq 3$, in European coordinates; (2) the mean Siluro-Devonian pole for stable Europe ($Q \geq 3$, Van der Voo 1990); and (3) the Lower Old Red Sandstone pole from Wales (this paper). Gondwana was positioned using the interpolated Siluro-Devonian pole for Gondwana given by Kent & Van der Voo (1990). (Reconstruction used Terra Mobilis™ software marketed by Denham & Scotese.)

Sediment source arguments (Soper & Woodcock 1990) and provenance links (Murphy & Hutton 1986) suggest that the first ('soft') closure of the Iapetus Ocean in Britain (collision of Laurentia and Eastern Avalonia) was Late Llandovery to Wenlock in age, with foreland basin subsidence (associated with collision related underthrusting) from Late Wenlock (Kneller 1991). The collision of Baltica and Laurentia (Scandian orogeny) is considered to have occurred in Wenlock/Ludlow time (McKerrow 1988; Torsvik *et al.* 1990). The similarity of fauna in southern Britain and Baltica from Late Ordovician is consistent with collision of Baltica and Eastern Avalonia and the closure of the intervening Tornquist Sea at this time (Cocks & Fortey 1982; Fortey & Cocks 1988).

Although the chronology of Ordovician/Silurian closure of the Iapetus Ocean remains unclear, there is now a substantial body of a palaeomagnetic data supporting the existence of an Early Devonian supercontinent, previously proposed on the basis of faunal provinciality (e.g. McKerrow & Ziegler 1972). It is only recently that available palaeomagnetic data have been consistent with this configuration (see Miller & Kent 1988; Kent & Van der Voo 1990). The new result from the Lower Old Red Sandstone gives an Early Devonian palaeolatitude of 17°S (±5°) for south Wales, which is consistent with coeval palaeolatitudes from Laurentia, Baltica and Armorica in a supercontinent configuration (Fig. 10).

ACKNOWLEDGMENTS

This research was supported by US National Science Foundation grants EAR-8816571 and EAR-8816802, and by NATO Scientific Affairs Division. We have benefited from discussions of this work with P. Turner, T. Torsvik and A. Trench.

REFERENCES

Almond, J., 1983. The sedimentology of the earliest Lower Old Red Sandstone, south central Wales, *PhD thesis*, University of Bristol.

Bachtadse V. & Briden, J. C., 1990. Palaeomagnetic constraints on the position of Gondwana during Ordovician to Devonian times, in *Palaeozoic Palaeogeography and Biogeography*, pp. 43–48, eds McKerrow, W. S. & Scotese, C. R., Geological Society of London.

Briden, J. C., 1970. Palaeomagnetic results from the Arrochar and Garabal Hill–Glen Fyne Igneous Complex, Scotland, *Geophys. J. R. astr. Soc.*, **21,** 457–470.

Briden, J. C. & Duff, B. A., 1981. Pre-Carboniferous palaeomagnetism of Europe north of the Alpine Orogenic belt, in *Palaeoreconstruction of the Continents*, pp. 137–149, eds McElhinny, M. W. & Valencio, D. A., American Geophysical Union, Washington, DC.

Briden, J. C., Turnell, H. B. & Watts, D. R., 1984. British paleomagnetism, Iapetus Ocean, and the Great Glen fault, *Geology*, **12,** 428–431.

British Geological Survey, 1988. *1:250,000 Geological Map, Bristol Channel.*

British Geological Survey, 1990. *1:250,000 Geologial Map, Mid Wales and Marches.*

Bullard, E. C., Everett, J. E. & Smith, A. G., 1965. A symposium on continental drift, IV, The fit of the continents around the Atlantic, *Phil. Trans. R. Soc. Lond.*, **258,** 41–51.

Bylund, G., 1974. Paleomagnetism of dykes along the southern margin of the Baltic Shield, *Geol. Foeren. Stockholm Foerh.*, **96,** 231–235.

Chamalaun, F. H., 1964. Origin of the secondary magnetization of the Old Red Sandstones of the Anglo-Welsh cuvette, *J. geophys. Res.*, **69,** 4327–4337.

Chamalaun, F. H. & Creer, K. M., 1964. Thermal demagnetisation studies on the Old Red Sandstone of the Anglo-Welsh cuvette, *J. geophys. Res.*, **69,** 1607–1616.

Channell, J. E. T., McCabe, C., Torsvik, T. H., Trench, A. & Woodcock, N. H., 1992. Recent advances in Palaeozoic palaeomagnetism of the Welsh basin, *Geol. Mag.*, submitted.

Claesson, K. C., 1978. Swedish Ordovician limestones: problems in clarifying their directions of magnetization, *Phys. Earth planet. Inter.*, **16,** 65–72.

Cocks L. R. M. & Fortey, R. A., 1982. Faunal evidence for oceanic separation in the Paleozoic for Britain, *J. geol. Soc. Lond.*, **139,** 465–478.

Creer, K. M., 1962. A statistical enquiry into the partial remagnetization of folded Old Red Sandstone rocks, *J. geophys. Res.*, **67,** 1899–1906.

Deutsch, E. R., 1980. Magnetism of the Mid-Ordovician Tramore volcanics, SE Ireland, and the question of a wide Proto-Atlantic Ocean, *J. Geomag. Geoelectr.*, **32,** III, 77–98.

Douglass, D. N., 1988. Paleomagnetics of Ringerike Old Red Sandstone and related rocks, southern Norway: implications for pre-Carboniferous separation of Baltica and British terranes, *Tectonophysics*, **148,** 11–27.

Esang, C. B. & Piper, J. D. A., 1984. Palaeomagnetism of the Carboniferous E–W dyke swarm in Argyllshire, *Scott. J. Geol.*, **20,** 309–314.

Fisher, R. A., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond.*, A, **217,** 295–305.

Fortey, R. A. & Cocks, L. R. M., 1988. Arenig to Llandovery faunal distributions in the Caledonides, in *The Caledonian–*

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Appalachian Orogen, Geological Society Special Publication No. 38, pp. 233–246, eds Harris, A. L. & Fettes, D. J., London.

Halvorsen, E., 1970. Paleomagnetism and age of the younger diabases in the Ny-Hellesund area, *S. Norway. Nor. Geol. Tidsskr.*, **50,** 157–166.

Hargraves, R. B., Dawson, E. M. & Van Houten, F. B., 1987. Palaeomagnetism and age of mid-Palaeozoic ring complexes in Niger, west Africa, and tectonic implications, *Geophys. J. R. astr. Soc.*, **90,** 705–729.

House, M. R., Richardson, J. B., Chaloner, W. G., Allen, J. R. L., Holland, C. H. & Westoll, T. S., 1977. *The Correlation of Devonian Rocks of the British Isles*, Special Report, 7, Geological Society, London.

Kelling, G., 1988. Silesian sedimentation and tectonics in the South Wales Basin: a brief review, in *Sedimentation in a Synorogenic Basin Complex: the Upper Carboniferous of Northwest Europe*, pp. 38–42, eds Besly, B. M. & Kelling, G., Blackie, Glasgow.

Kent, D. V. & Opdyke, N. D., 1985. Multicomponent magnetisations from the Mississippian Mauch Chunk Formation of the central Appalachians and their tectonic implications, *J. geophys. Res.*, **90,** 5371–5383.

Kent, D. V. & Van der Voo, R., 1990. Palaeozoic palaeogeography from palaeomagnetism of the Atlantic bordering continents, in *Palaeozoic Palaeogeography and Biogeography*, pp. 49–56, eds McKerrow W. S. & Scotese, C. R., Geological Society of London.

Kirschvink, J. L., 1980. The least squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. astr. Soc.*, **62,** 699–718.

Kneller, B. C., 1991. A foreland basin on the southern margin of Iapetus, *J. geol. Soc. Lond.*. **148,** 207–210.

Latham, A. G. & Briden, J. C., 1975. Paleomagnetic field directions in Siluro-Devonian lavas of the Lorne Plateau, Scotland, and their regional significance, *Geophys. J. R. astr. Soc.*, **43,** 243–252.

McCabe, C. & Elmore, R. D., 1989. The occurrence and origin of Late Palaeozoic remagnetisation in the sedimentary rocks of North America, *Rev. Geophys.*, **27,** 471–494.

McCabe, C. & Channell, J. E. T., 1990. Paleomagnetic results from volcanic rocks of the Shelve Inlier, Wales: evidence for a wide Late Ordovician Iapetus Ocean in Britain, *Earth planet. Sci. Lett.*, **96,** 458–468.

McCabe, C., Channell, J. E. T. & Woodcock, N. H., 1991. Further paleomagnetic results from the Builth Wells Ordovician Inlier, Wales, *J. geophys. Res.*, in press.

McClelland Brown, E., 1983. Palaeomagnetic studies of fold development and propagation in the Pembrokeshire Old Red Sandstone, *Tectonophysics*, **98,** 131–146.

McClelland, E., 1987. Palaeomagnetic results from the Devonian Llandstadwell Formation, Dyfed, Wales-Discussion, *Tectonophysics*, **143,** 335–336.

McElhinny, M. W., 1964. Statistical significance of the fold test in paleomagnetism, *Geophys. J. R. astr. Soc.*, **8,** 338–340.

McKerrow, W. S., 1988. The development of the Iapetus Ocean from the Arenig to the Wenlock, in *The Caledonian–Appalachian Orogen*, Special Publication, 38, pp. 405–412, eds Harris, A. L. & Fettis, D. J., Geological Society, London.

McKerrow, W. S. & Ziegler, A. M., 1972. Palaeozoic oceans, *Nature*, **240,** 92–94.

Miller, J. D. & Kent, D. V., 1986. Synfolding and prefolding magnetisations in the Upper Devonian Catskill formation of eastern Pennyslvania: implications for the tectonic history of Acadia, *J. geophys. Res.*, **91,** 12 791–12 803.

Miller, J. D. & Kent, D. V., 1986. Palaeomagnetism of the Siluro-Devonian Andreas redbeds: evidence for an Early Devonian supercontinent?, *Geology*, **16,** 195–198.

Morel, P. & Irving, E., 1978. Tentative paleocontinential maps for the Early Phanerozoic and Proterozoic, *J. Geol.*, **86,** 535–561.

Morris, W. A., Briden, J. C., Piper, J. D. A. & Sallomy, J. T., 1973. Palaeomagnetic studies in the British Caledonides—V, Miscellaneous new data, *Geophys. J. R. astr. Soc.*, **34,** 69–105.

Murphy, F. C. & Hutton, D. H. W., 1986. Is the Southern Uplands of Scotland really an accretionary prism?, *Geology*, **14,** 354–357.

Owen, T. R. & Weaver, J. D., 1983. The structure of the main South Wales Coalfield and its margins, in *The Variscan Fold Belt in the British Isles*, pp. 74–87, ed. Hancock, P. L., Adam Hilger, Bristol.

Perroud, H. & Bonhommet, N., 1981. Palaeomagnetism of the Ibero-Armorican arc and the Hercynian orogeny in western Europe, *Nature*, **292,** 445–448.

Perroud, H. & Van der Voo, R., 1985. Palaeomagnetism of the late Ordovician Thouars Massif, Vendee province, France, *J. geophys. Res.*, **90,** 4611–4625.

Perroud, H., Calza, F. & Khattach, D., 1991. Palaeomagnetism of the Silurian volcanism at Almaden, southern Spain, *J. geophys. Res.*, **96,** 1949–1962.

Piper, J. D. A., 1975. Palaeomagnetism of Silurian lavas of Somerset and Gloucestershire, England, *Earth planet. Sci. Lett.*, **25,** 355–360.

Piper, J. D. A., 1978. Palaeomagnetism and palaeogeography of the Southern Uplands block in Ordovician times, *Scott. J. Geol.*, **14,** 93–107.

Sallomy, J. T. & Piper, J. D. A., 1973. Palaeomagnetic studies in the British Caledonides—IV Lower Devonian lavas of the Strathmore Region, Scotland, *Geophys. J. R. astr. Soc.*, **34,** 47–68.

Schmidt, P. W., Embleton, B. J. J., Cudahy, T. J. & Powell, C. M., 1986. Prefolding and premegakinking magnetizations from the Devonian Comerong volcanics, New South Wales, Australia, and their bearing on the Gondwana pole path, *Tectonics*, **5,** 135–150.

Soper, N. J. & Woodcock, N. H., 1990. Silurian collision and sediment dispersal patterns in southern Britain, *Geol. Mag.*, **127,** 527–542.

Soper, N. J., Webb, B. C. & Woodcock, N. H., 1987. Late Caledonian transpression in northwest England: timing, geometry and geotectonic significance, *Proc. Yorks. Geol. Soc.*, **46,** 175–192.

Stearns, C. & Van der Voo, R., 1987. Palaeomagnetic results from the Lower Devonian Llandstadwell Formation, Dyfed, Wales, *Tectonophysics*, **143,** 329–334.

Storetvedt, K. M. & Torsvik, T. H., 1985. Paleomagnetic re-examination of the Middle–Upper Devonian Esha Ness ignimbrite, W. Shetland, *Phys. Earth planet. Inter.*, **37,** 169–173.

Torsvik, T. H., 1985a. Palaeomagnetic results from the Peterhead granite, Scotland; implication for regional late Caledonian magnetic overprinting, *Phys. Earth planet. Inter.*, **39,** 108–117.

Torsvik, T. H., 1985b. Magnetic properties of the Lower Old Red Sandstone lavas in the Midland Valley Scotland; paleomagnetic and tectonic considerations, *Phys. Earth planet. Inter.*, **39,** 194–207.

Torsvik, T. H. & Trench, A., 1991. The Lower-Middle Ordovician palaeofield of Scandanavia, southern Sweden 'revisited', *Phys. Earth planet. Inter.*, **65,** 283–291.

Torsvik, T. H., Olesen, O., Ryan, P. D. & Trench, A., 1990. On the palaeogeography of Baltica during the Palaeozoic: new palaeomagnetic data from the Scandanavian Caledonides, *Geophys. J. Int.*, **103,** 261–279.

Trench, A. & Torsvik, T. H., 1991. A revised Palaeozoic apparent polar wander path for southern Britain (Eastern Avalonia), *Geophys. J. Int.*, **104,** 227–233.

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Trench, A., Dentith, M. C., Bluck, B. J., Watts, D. R. & Floyd, J. D., 1989. Paleomagnetic constraints on the geological terrane models of the Scottish Caledonides, *J. geol. Soc. Lond.*, **146,** 405, 408.

Trench, A., Torsvik, T. H., Smethurst, M. A., Woodcock, N. H. & Metcalf, R., 1991. A palaeomagnetic study of the Builth Wells–Llandrindod Wells Ordovician Inlier, Wales: palaeogeographic and structural implications, *Geophys. J. Int.*, **105,** 477–489.

Trench, A., Torsvik, T. H., Dentith, M. C., Walderhaug, H. & Traynor, J.-J., 1992. A high southerly palaeolatitude for Southern Britain in Early Ordovician times: palaeomagnetic data from the Treffgarne Volcanic formation, SW Wales, *Geophys. J. Int.*, **108,** 89–100.

Turnell, H. B., 1985. Palaeomagnetism and Rb–Sr ages of the Ratagan and Comrie intrusions, *Geophys. J. R. astr. Soc.*, **83,** 363–384.

Turnell, H. B. & Briden, J. C., 1983. Palaeomagnetism of NW Scotland syenites in relation to local and regional tectonics, *Geophys. J. R. astr. Soc.*, **75,** 217–234.

Van der Voo, R., 1988. Palaeozoic palaeogeography of North America, Gondwana, and intervening displaced terranes: comparisons of palaeomagnetism with palaeoclimatology and biogeographical patterns, *Geol. Soc. Am. Bull.*, **100,** 311–324.

Van der Voo, R., 1990. Phanerozoic paleomagnetic poles from Europe and North America and comparisons with continential reconstructions. *Rev. Geophys.*, **28,** 167–206.

Van der Voo, R. & Johnson, R. J., 1985. Paleomagnetism of the Dunn Point Formation (Nova Scotia): high paleolatitudes for the Avalon Terrane in the Late Ordovician, *Geophys. Res. Lett.*, **12,** 337–340.

Watts, D. R., 1985. Palaeomagnetic resetting in the Barrovian zones of Scotland and its relationship to the late structural history, *Earth planet. Sci. Lett.*, **75,** 258–264.

Watts, D. R. & Briden, J. C., 1984. Palaeomagnetic signature of slow post-orogenic cooling of the north-east highlands of Scotland recorded in the Newer Gabbros of Aberdeenshire, *Geophys. J. R. astr. Soc.*, **77,** 775–788.

Woodcock, N. H., 1987. Structual geology of the Llandovery Series in the type area, Dyfed, Wales, *Geol. J.*, **22,** 199–209.

Woodcock, N. H., Awan, M. A., Johnson, T. E., Mackie, A. H. & Smith, R. D. A., 1988. Acadian tectonics of Wales during Avalonia/Laurentia convergence, *Tectonics*, **7,** 483–495.

Zijderveld, J. D. A., 1967. The natural remanent magnetization of the Exeter volcanic traps (Permian, Europe), *Tectonophysics*, **4,** 121–153.

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